Data Exploration and Analysis Pipeline

Thesis: Semi-Supervised Behavioral State Modeling with Dimensionality Reduction for Pediatric Digital Health Applications **Data:** National Survey of Children's Health (NSCH), topical and screener files, stored in DuckDB

Each phase lists its **steps**, the **decisions** you must justify, and the **thesis outputs** it produces. Variable names mentioned are examples to verify against the NSCH codebook for your survey year.

Phase 0. Project Foundations

Steps

1. Create one project folder with subfolders: `data_raw/`, `data_processed/`, `notebooks/`, `src/`, `figures/`, `tables/`, `docs/`.
2. Pin the environment (`pyproject.toml` and `uv.lock`). Record Python, DuckDB, scikit-learn, and UMAP versions.
3. Set a global random seed and record it. Use it in every stochastic step.
4. Keep raw data read-only. Every transformation is a script or notebook cell that can be re-run from raw data to final results.
5. Keep a research log (date, decision, reason). This becomes the audit trail for your methodology chapter.
6. Write the analysis plan before looking at outcomes: research questions, hypotheses, primary metrics, and the evaluation design. Freezing this early protects against post-hoc choices.

Thesis outputs: reproducibility statement, software and version table, analysis plan.

Phase 1. Data Acquisition and Provenance

Steps

1. Record the source URL, survey year, download date, and file names for each file.
2. Record checksums (SHA-256) of the downloaded zips.
3. Download the documentation that matches the data: Variable List, Frequencies, Analytic Guide, FAQs, Multi-Year Guide, and Imputation Guide.
4. Note the survey's sponsor, mode, sampling frame, response rates, and fielding period from the methodology documents.

Decisions: which years to use, and whether to pool. Pooling requires checking that every variable you use is coded consistently across years.

Thesis outputs: data source description, sampling design summary, a table of datasets used.

Phase 2. Ingestion and Structural Validation

Steps

1. Load files into DuckDB with a documented loader script. Preserve the raw numeric codes. Store variable labels and value labels in separate lookup tables (already done in your setup).
2. Verify row counts against the documentation's stated sample size.
3. Verify the column count against the Variable List.
4. Check data types. Many NSCH variables are stored as numbers but are categorical codes.
5. Check the unique identifier. Confirm there are no duplicate children (`COUNT(*) = COUNT(DISTINCT HHID)` or the equivalent key).
6. Decide how screener and topical files relate (join key, and whether you need the screener at all).
7. Reproduce at least two published weighted estimates from the Frequencies PDF. If your numbers match, your loading and weighting are correct.

Thesis outputs: validation table (your counts versus official counts), statement that the ingestion was verified.

Phase 3. Variable Dictionary and Taxonomy

Steps

1. Build a master data dictionary table: variable name, label, type (binary, nominal, ordinal, continuous), response options, valid range, missing and special codes, survey section, and applicable age range.
2. Assign each variable to a **domain**, for example: child demographics, physical health, emotional and behavioral health, developmental and learning, family functioning, parent health, adverse childhood experiences, neighborhood and community, health care access and use, sleep and screen time, school.
3. Flag **design variables** (weights, strata, primary sampling units), **identifiers**, and **administrative variables**. These must not be used as features.
4. Flag variables that are **conditionally asked** (skip patterns). A missing value may mean "not applicable," not "unknown."
5. Flag variables that are **mechanically related** (derived indicators computed from other items). They create redundancy and potential leakage.

Decisions: which domains support the concept of a "behavioral state," and which are context or confounders.

Thesis outputs: variable taxonomy table (appendix), conceptual framework figure linking domains to behavioral states.

Phase 4. Cohort Definition

Steps

1. Define the target population: age range, and any exclusions (for example, children too young for the behavioral items).
2. Define inclusion and exclusion criteria, and apply them one at a time while recording counts.
3. Produce a participant flow diagram showing the sample size remaining after each criterion.
4. Compare included and excluded children on key characteristics, to show whether exclusion

4. Compare included and excluded children on key characteristics, to show whether exclusion introduces bias.
5. Decide whether the analysis is on the whole population or a clinical subgroup (for example, children with any parent-reported mental health condition), and justify that scientifically.

Thesis outputs: flow diagram, inclusion and exclusion table, Table 1 (cohort characteristics).

Phase 5. Data Quality Audit

Steps

1. **Special codes.** Identify codes meaning "don't know," "refused," "not applicable," or "no valid response." Convert them to proper missing values with a documented mapping, keeping the reason category as a separate indicator where useful.
2. **Missingness profile.** Compute missing percentage per variable and per child. Plot a missingness matrix.
3. **Missingness mechanism.** Assess whether missingness depends on age, sex, or other variables (compare missing versus non-missing groups). Classify as plausibly MCAR, MAR, or MNAR, and state the assumption.
4. **Range and logic checks.** Values outside valid ranges, impossible combinations (for example, an age-inappropriate item answered), and inconsistencies between related items.
5. **Duplicates and near-duplicates.**
6. **Rare categories.** Variables with categories below a minimum frequency (for example, below 1% or 30 observations).
7. **Near-constant variables** (very low variance) and variables with excessive missingness (set and justify a threshold, for example above 40%).
8. **Outliers.** For continuous variables, use IQR or robust z-scores, but treat statistically extreme values as possibly valid before removing anything.

Decisions: missing data strategy (complete-case, imputation, or model-based handling), variable exclusion thresholds.

Thesis outputs: data quality table, missingness figure, documented thresholds.

Phase 6. Cleaning and Recoding

Steps

1. Apply the special-code mapping from Phase 5.
2. Recode variables to consistent direction (for example, higher always means more severe or more frequent).
3. Collapse sparse categories where theoretically defensible, and document each collapse.
4. Create the processed dataset as a new DuckDB table or Parquet file. Never overwrite raw tables.
5. Version the processed dataset (date and script hash).
6. Re-run the Phase 5 audit on the cleaned data to confirm the problems are resolved.

Thesis outputs: cleaning log, before and after summary.

Phase 7. Univariate Exploration

Steps

1. For every candidate feature: counts and percentages (categorical), or mean, median, standard deviation, skewness, and kurtosis (continuous).
2. Visualize: bar charts for categorical, histograms and density plots for continuous.
3. Compute both **unweighted** and **survey-weighted** distributions. Report weighted estimates when making statements about the US child population.
4. Examine ordinal items for ceiling and floor effects.
5. Tabulate each potential label variable (diagnoses, symptom indicators) with weighted prevalence.

Thesis outputs: descriptive statistics tables, prevalence table compared against published NSCH figures.

Phase 8. Bivariate and Multivariate Exploration

Steps

1. **Association measures appropriate to type:**
   - continuous with continuous: Spearman correlation
   - ordinal with ordinal: Spearman or Kendall
   - categorical with categorical: Cramér's V (bias-corrected) or mutual information
   - categorical with continuous: eta squared or point-biserial
2. Build association heatmaps within and across domains.
3. Identify **redundant clusters** of variables (high mutual association). This empirically justifies dimensionality reduction.
4. Check **multicollinearity** (variance inflation factors, where meaningful).
5. Explore how candidate labels relate to features, to confirm the features carry signal about the labels.
6. Explore subgroup differences (age band, sex, race and ethnicity, income band) for fairness considerations later.
7. Examine the data's intrinsic structure early with a quick PCA, MCA, or UMAP plot colored by labels. This is exploratory only. Do not tune the final model on it.

Thesis outputs: association heatmaps, redundancy analysis, exploratory embedding figure.

Phase 9. Survey Design Handling

Steps

1. Identify the weight, stratum, and PSU variables from the Analytic Guide.
2. Decide for each analysis whether it needs design-based inference:
   - **Descriptive prevalence and associations reported for the population:** use weights and design-based variance (Taylor linearization or replicate methods).

- **Machine learning model fitting:** state how weights are handled (unweighted fitting, weighted loss, or weighted evaluation) and justify the choice. Be explicit that cluster discovery on the sample is not the same as population estimation.

3. For multiply imputed variables (such as income and poverty measures), run analyses on each imputation and pool results using Rubin's rules. Follow the NSCH Imputed Data Guide.
4. If pooling years, apply the Multi-Year Guide (weight rescaling).

Thesis outputs: statement on how complex survey design was handled, listed as a methodological assumption or limitation.

Phase 10. Label Strategy and Semi-Supervised Design

Steps

1. Define the **labeled signal**. Options include parent-reported diagnosis of anxiety, depression, behavior or conduct problems, ADHD; a composite severity index; or a multi-class "state" derived from several indicators. Justify it clinically and cite literature.
2. Assess label **quality and limitations**: parent-reported, not clinician-verified, subject to reporting bias, and dependent on health care access for diagnosis. State this limitation explicitly.
3. Assess **class balance** and decide on handling (stratified sampling, class weights, or resampling applied only inside training folds).
4. Design the **label scarcity simulation.** Because NSCH has labels for most children, create "labeled" and "unlabeled" subsets by masking labels. For example, retain 1%, 5%, 10%, 25%, and 100% of labels, sampled in a stratified way and repeated across multiple random seeds.
5. Decide the **mechanism of masking:** random (MCAR labels) as the baseline, and optionally biased masking (labels more likely for severe cases) as a harder, more realistic test.
6. Lock the **test set** before any modeling, and never use it for tuning. Test labels are fully known and held out.

Thesis outputs: label definition table, label scarcity protocol diagram, justification of simulation validity.

Phase 11. Feature Engineering and Selection

Steps

1. Create domain composite scores where psychometrically justified (sum or mean scores of related items, with reliability measured by Cronbach's alpha or McDonald's omega).
2. Create interaction or derived features only when theoretically motivated.
3. **Remove leakage.** Exclude any feature that is the label, a recode of the label, or a direct consequence of it (for example, treatment variables when the label is the diagnosis).
4. Apply variable filtering: remove near-constant variables, those above the missingness threshold, and redundant ones (association above a threshold).
5. Decide on feature sets for comparison, for example: all eligible features; domain-restricted subsets (behavior only, behavior plus family context); and a clinically curated set.
6. Do any **data-driven selection inside cross-validation folds only**, to avoid optimistic bias.

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Thesis outputs: final feature list (appendix), feature set definitions, leakage exclusion table.

Phase 12. Preprocessing for Modeling

Steps

1. **Data splitting:** hold-out test set (for example, 20%), stratified by label and key demographics. Use the remaining data for training and validation with repeated stratified k-fold cross-validation.
2. **Fit all preprocessing on training data only,** then apply to validation and test. This includes imputation, scaling, encoding, and any variance filtering.
3. **Imputation:** choose and justify (mode or median, iterative imputation, or multiple imputation). Compare at least one alternative as a sensitivity analysis.
4. **Encoding:** one-hot for nominal variables, ordinal encoding or polychoric-based treatment for ordinal variables, standardization for continuous.
5. For mixed data, consider a distance suited to it (Gower) or an embedding that handles categorical inputs.
6. Save the fitted preprocessing pipeline as an object so the entire path from raw data to prediction is reproducible.

Thesis outputs: preprocessing pipeline diagram, justification of each choice.

Phase 13. Dimensionality Reduction

Steps

1. **Baselines:** no reduction (full feature space), and simple feature selection.
2. **Linear methods:** PCA (on appropriately encoded data), with scree plot, explained variance, and loadings interpretation. Consider Multiple Correspondence Analysis or categorical PCA for categorical data.
3. **Nonlinear methods:** UMAP, t-SNE (visualization only), kernel PCA, and autoencoders (including variational autoencoders).
4. **Semi-supervised or supervised reduction** where relevant, such as a semi-supervised autoencoder or supervised UMAP, to align with the framework's objectives.
5. **Choose the number of components** by explained variance, reconstruction error, downstream performance, or stability, and report the rule used.
6. **Evaluate the embeddings:**
   - reconstruction error
   - neighborhood preservation (trustworthiness and continuity)
   - stability across seeds and bootstrap resamples
   - downstream task performance
   - interpretability (which original variables drive each component)
7. Record **computational cost** (time and memory) for each method. This directly supports your objective of reducing computational overhead.

8. Fit the reducer on training data only, and project validation and test data through it.

Thesis outputs: method comparison table, scree and embedding figures, computational cost table.

Phase 14. Behavioral State Modeling

Steps

1. **Unsupervised baselines:** k-means, Gaussian mixture models, hierarchical clustering, latent class or latent profile analysis (the standard for survey data with categorical items), and optionally density-based clustering.
2. **Semi-supervised models**, for example:
   - label propagation and label spreading
   - self-training and co-training wrappers
   - semi-supervised Gaussian mixture models
   - semi-supervised deep models (a VAE with a classifier head, or consistency regularization methods)
3. **Supervised upper bound:** the same model trained with all labels. This sets the best-case reference.
4. **Fully unsupervised lower bound:** clustering without labels.
5. **Hyperparameter tuning** by nested cross-validation or a separate validation split, using only labeled validation data, and report the search space.
6. **Choosing the number of states:** BIC or AIC for mixture models, silhouette, stability analysis, and clinical interpretability. Justify the final number with both statistical and clinical evidence.
7. Run each configuration across multiple seeds and label-masking draws, and report means and confidence intervals.

Thesis outputs: model specification table, hyperparameter table, tuning protocol.

Phase 15. Validation

A. Statistical and predictive validation

1. Classification metrics on the held-out test set: balanced accuracy, macro F1, precision, recall, AUROC, AUPRC (important under class imbalance), and the confusion matrix.
2. Calibration (reliability curves, Brier score).
3. Cluster quality metrics: silhouette, Davies-Bouldin, and Calinski-Harabasz, plus agreement with known labels (adjusted Rand index, normalized mutual information).
4. Statistical comparison of methods using paired tests across folds or seeds, with multiple-comparison correction.

B. Stability validation

1. Bootstrap or subsample the data and measure how consistent the states are (adjusted Rand index between runs).
2. Compare solutions across random initializations.

C. Clinical and [illegible] validation

C. Clinical relevance validation

1. **Profile each state** with weighted prevalence of external variables not used to build the states (diagnoses, service use, school outcomes, family adversity, health care access).
2. Test whether states differ on these external variables (design-based tests, effect sizes).
3. Check **face validity** with clinical literature, and if possible, review with clinicians or supervisors who can judge whether the states are meaningful.
4. Name states cautiously and descriptively. State that they are data-driven profiles, not diagnoses.

D. Fairness validation

1. Compare performance and state distribution across sex, age band, race and ethnicity, and income. Report disparities.

Thesis outputs: results tables with confidence intervals, state profile table, stability figures, fairness table.

Phase 16. Robustness Across Data Conditions

This phase addresses your objective to evaluate the framework "across varying data conditions."

Experiments

1. **Label availability:** 1%, 5%, 10%, 25%, 50%, 100%. Plot performance against label fraction (learning curves) for each method.
2. **Sample size:** train on subsamples of the data.
3. **Feature dimensionality:** vary the number of input features or components.
4. **Noise and missingness:** inject additional missing values or label noise, and observe degradation.
5. **Class imbalance:** vary prevalence through resampling in the training set.
6. **Distribution shift:** train on one year or subgroup and test on another, where available.
7. **Ablations:** remove each framework component (dimensionality reduction, semi-supervision) in turn, to show each contribution.

Thesis outputs: learning curves, ablation table, sensitivity analysis section.

Phase 17. Interpretation

Steps

1. Global interpretation: component loadings, feature importance (permutation importance, SHAP where applicable), and centroid or profile plots per state.
2. Local interpretation for individual cases, as an illustration of potential digital health use.
3. Translate states into a narrative of how a digital health application could use them (monitoring, triage, tailoring content), while staying within what the data support.
4. Distinguish clearly between **association** and **causation**. Cross-sectional data cannot show how a child moves between states over time.

Thesis outputs: interpretation figures, application scenario section.

Phase 18. Reporting, Ethics, and Limitations

Steps

1. Follow a reporting standard (for example TRIPOD-style transparency for prediction models, adapted to unsupervised and semi-supervised work).
2. Confirm data use compliance: NSCH public use files are de-identified, but cite the data use terms and obtain any ethics determination your institution requires.
3. Write the limitations section covering:
   - cross-sectional design (no temporal dynamics)
   - parent-reported measures and label noise
   - simulated label scarcity versus real-world scarcity
   - complex survey design versus ML fitting
   - generalizability beyond the US child population
   - no direct behavioral sensor data, which a digital health deployment would add
4. Write future work: longitudinal data, wearable or app data, clinician-validated labels, prospective evaluation.
5. Archive code, environment, and the processed data specification so results are reproducible.

Deliverables Mapped to Thesis Chapters

Thesis section	Pipeline phases
Data and ethics	1, 2, 18
Methodology: data preparation	3, 4, 5, 6, 9, 12
Exploratory data analysis results	7, 8
Methodology: framework design	10, 11, 13, 14
Results: model performance	15A, 15B, 16
Results: clinical relevance	15C, 15D, 17
Discussion and limitations	17, 18

Order of Operations Checklist

1. Freeze the analysis plan.
2. Ingest and verify the data against official figures.
3. Build the data dictionary and cohort.
4. Audit and clean the data.
5. Explore, both unweighted and weighted.
6. Define labels and the label scarcity protocol, and lock the test set.
7. Engineer and select features without leakage.

8. Build the preprocessing pipeline inside the cross-validation structure.
9. Compare dimensionality reduction methods.
10. Fit unsupervised, semi-supervised, and supervised models.
11. Validate statistically, then for stability, then clinically, then for fairness.
12. Stress-test across data conditions.
13. Interpret, document limitations, and archive.